

VM Setup Guide

AWS EC2 — Ubuntu 24.04 LTS — Production-Ready Configuration

1. Cloud Provider and Instance Selection

Provider: Amazon Web Services (AWS) — EC2

AWS was selected for its reliability, immediate availability, and student credit program. The instance is covered by AWS credits during the development period.

Instance Specifications

Parameter	Value
Instance Type	t3.small
vCPUs	2
RAM	2 GiB
Architecture	x86_64 (amd64)
Operating System	Ubuntu Server 24.04 LTS
AMI	Canonical, Ubuntu, 24.04, amd64 noble image
Storage	30 GiB gp3 EBS (3000 IOPS)
Region	US East (N. Virginia) — us-east-1
Network	Default VPC, public subnet, auto-assigned public IPv4

Cost

Case 1 — Standard pricing (no credits or benefits)

Item	Rate	Monthly Estimate
t3.small compute	\$0.0074 / hour	~\$5.40 / month
30 GiB gp3 EBS storage	\$0.08 / GiB / month	~\$2.40 / month
Data transfer (first 100 GB)	Free	\$0.00
Total — standard		~\$7.80 / month

Case 2 — Current situation: AWS Free Plan with student credits

Item	Rate	Monthly Estimate	Covered by
t3.small compute	\$0.0074 / hour	~\$5.40 / month	AWS credits
30 GiB gp3 EBS storage	\$0.08 / GiB / month	~\$2.40 / month	AWS credits

Data transfer (first 100 GB)	Free	\$0.00	Always free
Out-of-pocket cost		\$0.00 / month	Until Aug 5, 2026

Account details: AWS Free Plan activated February 5, 2026. Credits available: \$117.21 USD. All compute and storage costs are deducted from this credit balance — no charges are billed to a payment method while credits remain.

Credit expiration: August 5, 2026 (or when credits are fully depleted, whichever comes first). After this date the account upgrades automatically to a standard paid plan and charges apply at the rates shown in Case 1 (~\$7.80/month). To reduce ongoing cost after expiration, downgrade the instance to t3.micro (1 GiB RAM, \$0.0036/hour, ~\$2.60/month compute) and reduce the EBS volume to 8 GiB (~\$0.64/month), bringing the total to ~\$3.24/month.

2. Security Group (Firewall)

A new security group was created during instance launch with the following inbound rule:

Type	Protocol	Port	Source	Purpose
SSH	TCP	22	0.0.0.0/0	Remote terminal access

No HTTP/HTTPS ports were opened on the security group. Application traffic is routed exclusively through Tailscale, which does not require any inbound ports.

3. SSH Key Pair and Access

An ED25519 key pair was generated at instance creation time. The private key was downloaded and stored locally. The default login user for Ubuntu AMIs on AWS is ubuntu.

Key setup (run once on local machine):

```
chmod 600 ~/Desktop/running_dashboard/keys/aws-app-server-key.pem
```

Connect command:

```
ssh -i ~/Desktop/running_dashboard/keys/aws-app-server-key.pem ubuntu@<PUBLIC_IP>
```

The public IP is visible in EC2 > Instances in the AWS console. After Tailscale is configured, the VM can also be reached via its Tailscale IP (100.121.195.112) without specifying the key file, as long as Tailscale is active on both devices.

4. Base System Setup

4.1 Update and upgrade system packages

```
sudo apt update && sudo apt upgrade -y
```

4.2 Install core dependencies

```
sudo apt install -y python3-pip python3-venv pipx git sqlite3 build-essential
```

Packages installed:

Package	Purpose
python3-pip	Python package installer
python3-venv	Python virtual environment support
pipx	Install Python CLI tools in isolated environments
git	Version control
sqlite3	Lightweight embedded database
build-essential	C/C++ compilers and build tools (required by some Python packages)

4.3 Configure pipx PATH

```
pipx ensurepath
```

After running pipx ensurepath, log out and back in (or run source ~/.bashrc) so the updated PATH takes effect.

5. Tailscale — Private Network

Tailscale creates an encrypted peer-to-peer VPN between all authorized devices. Once configured, the VM is reachable from any device on the same Tailscale network without exposing additional ports to the internet.

5.1 Install Tailscale on the VM

```
curl -fsSL https://tailscale.com/install.sh | sh
```

5.2 Authenticate the VM to the Tailscale network

```
sudo tailscale up
```

This prints a login URL. Open it in a browser on any device already signed in to the Tailscale account to authorize the VM. Tailscale is configured as a systemd service and starts automatically on reboot — no manual intervention required.

Network topology after setup:

Device	Tailscale IP	Status
janiels-macbook-air	100.77.136.64	Connected
ip-172-31-16-169 (AWS VM)	100.121.195.112	Connected

To connect to the VM via SSH using Tailscale (no key file needed once on the same Tailscale network if SSH keys are configured — standard key auth still applies):

```
ssh -i ~/Desktop/running_dashboard/keys/aws-app-server-key.pem  
ubuntu@100.121.195.112
```

6. Node.js Installation

Node.js is required to build the React frontend. It is installed from the official NodeSource repository to ensure an up-to-date LTS version, as the version available via apt is typically outdated.

```
curl -fsSL https://deb.nodesource.com/setup_lts.x | sudo -E bash -
```

```
sudo apt install -y nodejs
```

```
node -v
```

7. Final Verification

After completing all steps, the following versions were confirmed on the VM:

```
python3 --version && node -v && git --version && sqlite3 --version
```

Tool	Version Confirmed
Python	3.12.3
Node.js	v24.17.0
Git	2.43.0
SQLite3	3.45.1 (2024-01-30)

Full output from verification command:

```
Python 3.12.3 v24.17.0 git version 2.43.0 3.45.1 2024-01-30 16:01:20
e876e51a0ed5c5b3126f52e532044363a014bc594cfefa87ffb5b82257cc (64-bit)
ubuntu@ip-172-31-16-169:~$
```

8. File Transfer to VM

Files and project folders can be transferred from the local machine to the VM using scp or rsync over the Tailscale network.

Copy a single file:

```
scp -i ~/Desktop/running_dashboard/keys/aws-app-server-key.pem /path/to/file
ubuntu@100.121.195.112:~/
```

Copy a folder (recommended for project deployment):

```
rsync -avz -e "ssh -i ~/Desktop/running_dashboard/keys/aws-app-server-key.pem"
/path/to/project/ ubuntu@100.121.195.112:~/project/
```

rsync only transfers files that have changed, making subsequent syncs fast. The trailing slash on the source path is significant — it copies the contents of the folder rather than the folder itself.